OpenSK ML-DSA-87 Three-Way Correlated Log Analysis

Conversation: Compile and Install OpenSK (feadf5d5-e6b8-4687-ae9b-04083cd8b306)

Here is the three-way correlated analysis across the **Total Phase Beagle USB 12 Protocol Analyzer** capture (`/tmp/totalphase.csv`), the **Chrome Canary FIDO device log** (`/tmp/chrome_device.log` / `chrome_fido.log`), and the **OpenSK nRF52840 RTT debug log** (`rtt.log`).

1. Capture Window & USB Device Enumeration

- **Total Phase Capture (`/tmp/totalphase.csv`)**:
  - Recorded on `Dev=01, Ep=02` (USB Full-Speed 64-byte HID Interrupt IN/OUT endpoints) starting at **`21:28:55`** (`0:00.000.000`) through **`21:30:24`** (`1:28.917.984`).
- **Chrome Device Log (`chrome_fido.log`)**:
  - `21:06:17.987` : Total Phase analyzer enumerated (`vendor=5753 "Total Phase, Inc.", product=8193 "Beagle USB 12 Protocol Analyzer", serial="TP1112-175360"`).
  - `21:26:48.383` : Newly flashed OpenSK device enumerated (`vendor=6353 "Google Inc.", product=569 "Wasefire"`) with `max_msg_size = 16384` (`0x4000`).

2. Event 1 — Provisioning `ML-DSA-87` (`-50`) Batch Attestation (`7,678` Bytes, Host → OpenSK `SEQ` Rollover)

`tools/configure.py` provisioned `crypto_data_mldsa87/opensk_cert.pem` (`7,614` -byte DER certificate), `crypto_data_mldsa87/opensk.key` (`32` -byte seed), and `aaguid.txt` (`deadbeef-0000-0000-0000-deadbeef0000`) on CID `00 00 00 01` :

Timestamp (Beagle USB)	OpenSK RTT (`rtt.log`)	Beagle CSV (`totalphase.csv`)	Direction & Frame Details
`0:13.852.770`	`1564.523s` (L242)	L15–L16 (`Idx 15/20`)	`CTAPHID_INIT` (`FF FF FF FF 86 00 08 CE A8 D9 EE 29 0E 32 09`) → allocates CID **`00 00 00 01`** .
`0:13.864.771` – `0:14.059.795`	`1564.535s` (L246)	L17–L25 (`Idx 26–67`)	`AuthenticatorGetInfo` (`0x04`) → `457` B (`0x01C9`) response. In Idx 52 (L22): `05 19 40 00` confirms **`maxMsgSize: 16384 ( 0x4000 )`** and unconfigured `aaguid: 0000...0000` .
`0:14.064.796`	`1564.737s` (L260)	L26–L27 (`Idx 72/77`)	`CTAPHID_WINK` (`00 00 00 01 88 00 00`).
`0:14.304.825`	`1565.502s` (L278)	L37 (`Idx 129` OUT)	**Vendor Configure `INIT` packet**: `00 00 00 01 90 1D FE 40 A2 01 50 DE AD BE EF...` (`CMD=0x90` , **`BCNT=0x1DFE = 7,678` bytes**, cmd `0x40`).
`0:14.308.826` – `0:14.816.888`	—	L38–L165 (`Idx 134–769` OUT)	`CONT` packets `SEQ = 0x00` through `SEQ = 0x7F` (`128` packets, `4 ms` apart) carrying the DER X.509v3 certificate (`OID 2.16.840.1.101.3.4.3.19` / `id-ml-dsa-87` , `2,592` -byte public key, `4,627` -byte signature).
`0:14.820.889`	—	**L166 (`Idx 774` OUT)**	**Modulo-128 `SEQ` Rollover on Host → OpenSK!** `00 00 00 01 00 40 54 14...` (`SEQ` wraps from `0x7F` → **`0x00`** on the 129th CONT packet).
`0:14.824.889`	`1565.502s` (L279–L324)	L167 (`Idx 779` OUT)	Final `CONT` packet (`SEQ = 0x01` , 130th CONT packet): `...0C 5A 36 03 38 31 00...` ending with CBOR `03: 38 31` (`alg: -50` / `ML-DSA-87`).
`0:16.824.185`	`1567.493s` (L325–L327)	L169 (`Idx 785` IN)	**Configure Response**: `00 00 00 01 90 00 08 00 A3 01 F5 02 F5 03 F5` (`CTAP2_OK + {1: true, 2: true, 3: true}` — AAGUID, Certificate, and Private Key stored in flash).

3. Event 2 — Pure ML-DSA-87 (-50) MakeCredential with ML-DSA-87 (-50) Batch Attestation (15,176 Bytes, 2× SEQ Rollover!)

Approximately 20 seconds after provisioning (21:29:32 wall-clock / 0:36.735 Beagle time), Chrome Canary initiated a WebAuthn registration on demo.yubico.com over CID 00 00 00 02:

Phase A: GetInfo & MakeCredential Request

Beagle USB (_totalphase.csv)	Chrome Log (_chrome_fido.log)	OpenSK RTT (_rtt.log)	Event Details
0:36.735.576 (L179–L180)	21:29:32.122 (L101)	1587.405s (L328)	CTAPHID_INIT nonce DC A3 FD 5B B4 33 9F 0D → allocates CID **00 00 00 02**.
0:36.747.578 – 0:36.954.603 (L181–L189)	21:29:32.141 – 21:29:32.366 (L102–L117)	1587.419s (L332–L344)	AuthenticatorGetInfo (0x04) returns newly programmed aaguid: h'DEADBEEF000000000000DEADBEEF0000', maxMsgSize: 16384, and algorithms [-50, -49, -48, -8, -53, -7, -35, -36].
0:36.959.604 – 0:36.971.605 (L190–L193)	21:29:32.366 (L118)	1587.811s (L345–L373)	AuthenticatorMakeCredential (0x01, 226 B across 1 INIT + 3 CONT frames) for demo.yubico.com requesting pubKeyCredParams: [{alg: -50}, {alg: -8}, {alg: -7}].

Phase B: Dual ML-DSA-87 Execution on nRF52840 (2.39 s Crypto Compute)

- In _rtt.log (L346–L374):
  - 1587.811s : AuthenticatorMakeCredential command received.
  - 1590.200s : Sending response: AuthenticatorMakeCredential(...) ready.
  - **Total cryptographic execution time = 2.39 seconds** on the 64 MHz Cortex-M4F to perform **two** ML-DSA-87 (-50) operations back-to-back:
    1. Generate a new **ML-DSA-87 credential keypair** (2,592-byte public key, wrapped seed in 241-byte credential ID), and
    2. Sign authData || clientDataHash with the **ML-DSA-87 batch attestation key** (4,627-byte signature).
  - *(From 1590.200s to 1596.607s, the debug build printed the 15.2 KB response struct over SWD RTT before starting USB transmission at 0:46.591.837.)*

Phase C: 15,176-Byte (0x3B48) USB Transfer & Double Modulo-128 SEQ Rollover

The MakeCredential response is **15,176 bytes (0x3B48)** — nearly **2×** the standard 7,609-byte CTAP2 limit:

- fmt : "packed"
- authData : **2,898 bytes (0x0B52)** (rpIdHash + flags 0x45 + counter 1 + aaguid: deadbeef... + 241-byte wrapped credential ID + COSE_Key {1: 7 (AKP), 3: -50 (ML-DSA-87), -1: <2592-byte pubkey>})
- attStmt :
  - alg : **-50 (ML-DSA-87, 38 31)**
  - sig : **4,627 bytes (0x1213)** (0e dc 63 b1 50 95...)
  - x5c : **[7,614 bytes (0x1DBE)]** (30 82 1d ba..., the provisioned ML-DSA-87 batch X.509v3 certificate)

On the USB wire (_totalphase.csv L198–L455), OpenSK transmitted all **15,176 bytes** in **258 USB HID packets** (1 INIT + 257 CONT packets, 1 packet every 4.0 ms = **1.028 seconds** total wire transfer time), rolling over the 7-bit SEQ counter **twice**:

Beagle USB Time	CSV Line & Index	Packet Type & SEQ	Payload Milestone
0:46.591.837	L198 (Idx 894 IN)	INIT (CMD = 0x90, **BCNT = 0x3B48** = 15,176 B)	00 00 00 02 90 3B 48 00 A3 01 66 70 61 63 6B 65 64 02 59 0B 52... (status=0x00, "packed", start of 2,898-byte authData).
0:46.595.786 – 0:46.783.809	L199–L246 (Idx 900–1135 IN)	CONT SEQ = 0x00 .. 0x2F	2,592-byte ML-DSA-87 credential public key inside authData; at Idx 1135 (SEQ=0x2F, L246), attStmt starts: 03 A3 63 61 6C 67 38 31 63 73 69 67 59 12 13 0E DC 63 B1... ("alg": -50, "sig": <4627 bytes>).
0:47.103.849	L326 (Idx 1535 IN)	CONT **SEQ = 0x7F** (128th CONT)	End of 4,627-byte ML-DSA-87 attestation signature (...0B 13 1D 26) + start of "x5c" : 63 78 35 63 81 59 1D BE 30 82 1D BA... (7,614-byte ML-DSA-87 cert!).
0:47.107.849	**L327 (Idx 1540 IN)**	CONT **SEQ = 0x00** (129th CONT)	**1st Modulo-128 SEQ Rollover (0x7F → 0x00)** on OpenSK → Host!
0:47.107.849 – 0:47.615.912	L327–L454 (Idx 1540–2175 IN)	CONT SEQ = 0x00 .. 0x7F (129th–256th CONT)	Full body of the 7,614-byte ML-DSA-87 X.509v3 certificate (OID 2.16.840.1.101.3.4.3.19).
0:47.619.912	**L455 (Idx 2180 IN)**	CONT **SEQ = 0x00** (257th CONT)	**2nd Modulo-128 SEQ Rollover (0x7F → 0x00)** on OpenSK → Host! Final 15 bytes of the ML-DSA-87 cert (9F A7 B2 E2 00 00 00 07 13 1D 26 2C 34 3B 48) + zero padding.

Phase D: Chrome Canary Reassembly & Acceptance

At **21:29:43.030** (_chrome_fido.log L119–L120), immediately after Idx 2180 (0:47.619.912) arrived on the USB bus:

```
[21:29:43.030] EVENT ctap2_device_operation.h:186 -> {1: "packed", 2: (2898 bytes), 3: "[redacted]"}
[21:29:43.030] DEBUG authenticator_request_dialog_model.cc:158 UI step: kClosed
```

Chrome Canary seamlessly reassembled all **258 USB HID frames (15,176 bytes)** across **both** SEQ wrap-arounds, parsed the 2,898-byte ML-DSA-87 authData and the 12,262-byte ML-DSA-87 attStmt (4,627-byte signature + 7,614-byte ML-DSA-87 x5c certificate), and completed the registration ceremony (UI step: kClosed).